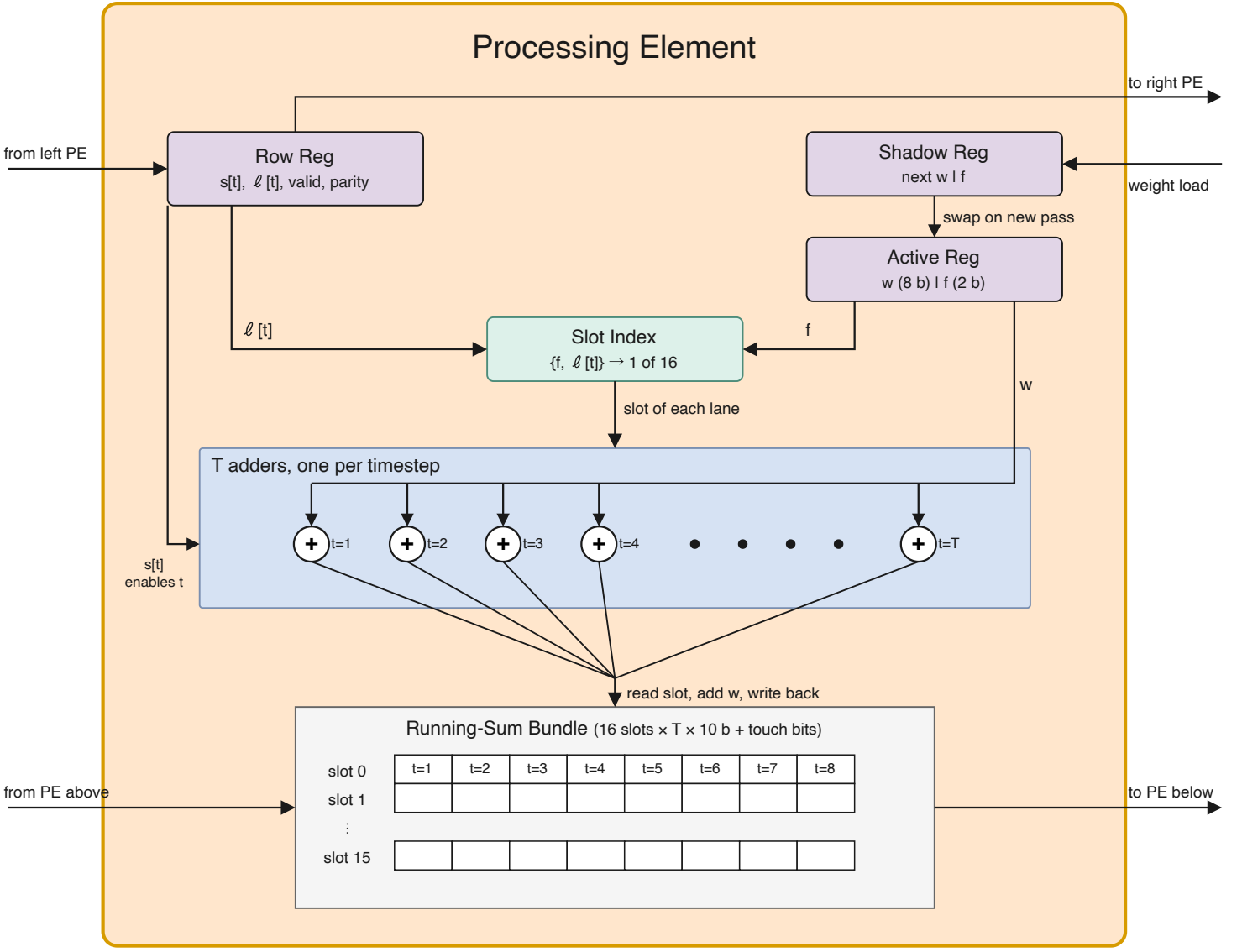
Processing Element
to right PE
from left PE
Row Reg
s[t], ℓ[t], valid, parity
Shadow Reg
next w l f
weight load
swap on new pass
Active Reg
w (8 b) l f (2 b)
ℓ[t]
Slot Index
{f, ℓ[t]} → 1 of 16
f
w
slot of each lane
T adders, one per timestep
t=1
t=2
t=3
t=4
t=T
s[t]
enables t
read slot, add w, write back
Running-Sum Bundle (16 slots × T × 10 b + touch bits)
slot 0
slot 1
slot 15
t=1
t=2
t=3
t=4
t=5
t=6
t=7
t=8
from PE above
to PE below